

FINAL EXPLANATION: BIOLOGY GRADE 11

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 11 – SUB-STRAND 1.2 ECOLOGY: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving Question: Why are fish disappearing from a water body that once supported many fish, and what can the community do to restore the ecosystem? Use this document as a model for your own Final Explanation. Remember the river video from Lesson 1: fewer fish, but more algae and weeds, and a bad smell. Write in your own words, use evidence from your investigations, readings, presentations and drawings (Lessons 2 to 9), and make sure your explanation explains ALL three puzzling observations. Your explanation should link ecosystem interactions, adaptations, populations, nutrient cycles and pollution, and end with realistic community actions. Use Kenyan examples such as Lake Victoria, the Nairobi River, Kisumu, Kericho and Rift Valley farms.

Part 1: The River as an Ecosystem (Biotic and Abiotic Factors)

Describe the river as an ecosystem. Name the biotic and abiotic factors and explain how they interact. Then explain why the idea that fish are old or have no food does not explain what you saw in the video.

The river is an ecosystem. Living (biotic) factors include fish like tilapia, algae, water weeds, insects and bacteria. Non-living (abiotic) factors include water, light, temperature, oxygen dissolved in water, and nutrients in the soil and water. These parts constantly interact: algae use light and nutrients to make food, small animals eat algae, fish eat those animals, and bacteria break down dead matter. In the video, the idea that the fish are just old cannot explain why algae and weeds are booming or why the water smells. The idea of no food also fails, because there is plenty of plant growth. So something in the environment has changed, and it is affecting how the organisms and their surroundings interact. When one factor changes, such as nutrients or oxygen, the whole ecosystem can change. That is why we must look at the river as a connected system, not just at the fish alone.

Part 2: Adaptations and Why Organisms Survive Only in Certain Habitats

Explain how adaptations help organisms survive in particular habitats. Use examples from your Lesson 2 presentation (for example hydrophytes and xerophytes) and explain what happens when a habitat changes faster than organisms

Environments are very different across Kenya, from the dry lands of Turkana to the wet shores of Lake Victoria, so organisms cannot survive everywhere. Each one has adaptations that suit its habitat. Xerophytes like cacti and sisal have thick waxy cuticles and rolled or reduced leaves to save water. Hydrophytes like water lilies have wide flat floating leaves and air spaces in their tissues so they can live in water. Fish have gills to take oxygen from water, streamlined bodies to swim, and scales to protect them. Fish are adapted to water with enough dissolved oxygen. If the river changes quickly, for example when oxygen drops, the fish cannot adapt fast enough because adaptations take many generations to develop. Fish then die or move away. Meanwhile, organisms that tolerate rich, low-oxygen water, such as some algae and floating weeds like water hyacinth, spread quickly. This helps explain why the fish decline while weeds increase.

can cope.	
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Part 3: Populations, Sampling and Evidence of Change

Explain what a population is and how scientists measure population size and density. Describe how you could show that the fish population is falling and the weed population is rising in the river.	A population is all the organisms of the same species living in the same area, such as all the tilapia in one stretch of river. Population size is how many there are, and density is how many per area or volume. These measures tell us how healthy a habitat is. For small, non-moving organisms, like weeds on the school compound, we mark a boundary such as a 10 m x 10 m grid and count every individual. For large or moving populations, like fish, we cannot count every one, so we use sampling methods such as quadrats for plants and capture-mark-recapture for fish. In capture-mark-recapture, we catch fish, mark them, release them, then catch another sample. $\text{Population} = (\text{number first marked} \times \text{total second catch}) / \text{number of marked fish recaptured}$. If we repeat this each year, falling fish numbers and rising weed density in the same stretch would give strong evidence of the change. Fishers' catch records from Kisumu or Lake Victoria can also support this evidence.
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Part 4: Nutrient Cycles and Pollution – Explaining the Algae, Weeds and Smell

Use your nitrogen and carbon cycle drawings (Lesson 8) and your summary of pollution effects (Lesson 9) to explain why algae and weeds are booming, why the water smells, and why fish are disappearing. Link each observation to a cause.	In the nitrogen cycle, nitrogen moves between air, soil, plants, animals and decomposers. Bacteria change nitrogen into nitrates that plants absorb. In the carbon cycle, plants and algae take in carbon dioxide in photosynthesis, and organisms release it by respiration and decay. When fertilisers from farms, sewage and detergents from towns, and waste from homes wash into the river, too many nitrates and phosphates enter the water. This is called eutrophication. Algae and weeds grow explosively. When they die, bacteria decompose them and use up lots of dissolved oxygen. Low oxygen kills fish, especially young ones. Decay without enough oxygen produces gases that cause the unpleasant smell. Thick weed and algae mats also block light so submerged plants die, and food chains collapse. So the three puzzles share one cause: pollution adding extra nutrients. Other pollution, like chemicals and plastics, can poison fish directly. Overfishing can make the problem worse.
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Part 5: What the Community Can Do to Restore the Ecosystem

Propose a realistic action plan for your community. Explain how each action fixes a cause you identified, and how you would check if it is working.	First, reduce nutrients entering the river: use fertiliser carefully, plant napier grass and trees along the riverbank as buffers, treat sewage, and stop dumping waste and washing with detergents in the river. Second, remove excess weeds by hand or with machines, and use the weeds for compost or biogas so nutrients leave the water. Third, follow fishing rules, such as correct net sizes and closed seasons, so fish can breed and populations recover. Fourth, educate neighbours through chiefs' barazas, schools and community groups, and work with NEMA and county officers. To check progress, measure the same things over time: fish population using capture-mark-recapture, weed density using quadrats, water smell and clarity, and dissolved oxygen. If oxygen rises, weeds shrink and fish return, the actions are working. If not, we revise our plan. This shows that people caused the problem and people can solve it by understanding how organisms interact with their environment.
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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Ecosystem interactions and	Clearly explains biotic and abiotic factors	Names biotic and abiotic factors and	Lists some factors or adaptations but

adaptations	and how they interact, uses accurate adaptation examples, and links habitat change to survival of fish and weeds.	gives some correct adaptation examples, with limited links to the river phenomenon.	does not explain interactions or connect them to the river.
Populations and sampling methods	Accurately defines population, size and density, and correctly chooses and explains counting or sampling methods (quadrat, capture-mark-recapture) with how they give evidence of change.	Defines population and describes at least one sampling method with minor gaps.	Confuses population terms or does not describe how populations are measured.
Explanation of the phenomenon using nutrient cycles and pollution	Uses cycle drawings and pollution evidence to explain algae and weed growth, smell and fish loss as one connected cause-and-effect chain, including eutrophication and low oxygen.	Explains most observations with mostly correct causes but misses one link in the chain.	Describes pollution in general terms without explaining the algae, smell or fish decline.
Use of evidence and Kenyan context	Draws on work from Lessons 2 to 9 (presentations, reading summary, drawings) and relevant Kenyan examples to support every claim.	Uses some evidence from lessons and a few Kenyan examples.	Gives claims with little or no evidence or local context.
Community solutions and communication	Proposes realistic, specific actions linked to causes, explains how to monitor success, and writes clearly in an organised way.	Proposes reasonable actions with some link to causes and some monitoring ideas, mostly clear writing.	Suggests vague actions not linked to causes, with unclear writing.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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